

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{18} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,0}[m_2] + \frac{1}{12} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,-1}[m_2] - \frac{4}{45} g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[m_2] - \right. \\
& \frac{1}{216} \sum_p g_1^6 \text{LF}_{3,0}[m_d^p] + \frac{1}{12} g_1^4 \overline{y_d^{pr}} y_d^{pr} \text{LF}_{3,0}[m_d^r] - \\
& \frac{1}{24} \sum_p g_1^6 \text{LF}_{3,0}[m_e^p] + \frac{1}{4} g_1^4 \overline{y_e^{pr}} y_e^{pr} \text{LF}_{3,0}[m_e^r] + \\
& \frac{1}{288} (18 \overline{y_e^{pr}} y_e^{pr} (g_1^4 + 2 g_1^2 g_2^2 + 5 g_2^4) + \sum_p (3 g_1^6 - g_1^2 g_2^4 - 10 g_2^6)) \text{LF}_{3,0}[m_l^p] + \\
& \frac{1}{96} g_2^2 (g_1^2 + g_2^2) (-24 \overline{y_e^{pr}} y_e^{pr} + \sum_p g_2^2) \text{LF}_{4,-1}[m_l^p] + \frac{1}{45} \sum_p g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[m_l^p] + \\
& \frac{1}{864} (18 \overline{y_d^{pr}} y_d^{pr} (g_1^4 + 42 g_1^2 g_2^2 + 45 g_2^4) - \sum_p (g_1^6 + 117 g_1^2 g_2^4 + 90 g_2^6)) \text{LF}_{3,0}[m_q^p] + \\
& \frac{1}{32} g_2^2 (g_1^2 + g_2^2) (-24 \overline{y_d^{pr}} y_d^{pr} + \sum_p g_2^2) \text{LF}_{4,-1}[m_q^p] + \frac{1}{15} \sum_p g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[m_q^p] + \\
& \frac{1}{27} \sum_p g_1^6 \text{LF}_{3,0}[m_u^p] + \frac{1}{36} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,0}[\tilde{\mu}] + \frac{1}{24} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,-1}[\tilde{\mu}] - \\
& \frac{2}{45} g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[\tilde{\mu}] - \frac{1}{12} g_1^6 \text{LF}_{3,3,-3}[m_1, \tilde{\mu}] - \frac{1}{4} g_1^6 m_1^2 \text{LF}_{3,3,-2}[m_1, \tilde{\mu}] - \\
& \frac{1}{3} g_2^4 (g_1^2 + g_2^2) \text{LF}_{2,1,0}[m_2, \tilde{\mu}] - \frac{17}{24} g_2^4 (g_1^2 + g_2^2) \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{6} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \frac{7}{6} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{3}{4} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] - \frac{1}{4} g_2^4 (2 g_1^2 + 5 g_2^2) \text{LF}_{3,3,-3}[m_2, \tilde{\mu}] + \\
& \frac{1}{4} g_2^4 m_2^2 (2 g_1^2 + g_2^2) \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-3}[m_2, \tilde{\mu}] + \\
& \frac{1}{2} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[m_2, \tilde{\mu}] - \frac{1}{2} g_1^2 \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} \text{LF}_{2,1,0}[m_d^r, m_d^t] + \\
& \frac{1}{12} g_1^2 \overline{a_d^{pr}} a_d^{pr} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,2,0}[m_d^r, m_q^p] + \frac{1}{12} g_1^4 \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,1,0}[m_d^r, m_q^p] - \\
& \frac{1}{2} g_1^2 \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} \text{LF}_{2,1,0}[m_e^r, m_e^t] + \frac{1}{4} g_1^2 g_2^2 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{2,2,0}[m_e^r, m_l^p] + \\
& \frac{1}{4} g_1^4 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,1,0}[m_e^r, m_l^p] + \frac{1}{4} g_2^4 \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,1,0}[m_l^p, m_e^r] - \\
& \frac{1}{8} g_1^2 \overline{a_e^{pr}} a_e^{pr} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_l^p, m_e^r] - \frac{1}{8} g_2^2 \overline{a_e^{pr}} a_e^{pr} (g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_l^p, m_e^r] + \\
& \frac{1}{24} \overline{a_e^{pr}} a_e^{pr} (3 g_1^2 - g_2^2) (g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_l^p, m_e^r] - \\
& \frac{1}{2} g_2^2 \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} \text{LF}_{2,1,0}[m_l^p, m_l^s] + \frac{1}{4} \overline{y_e^{pr}} \overline{y_e^{st}} y_e^{pt} y_e^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_l^p, m_l^s] + \\
& \frac{1}{12} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,1,0}[m_q^p, m_d^r] (2 g_1^2 + 3 g_2^2)^2 - \frac{1}{8} g_1^2 \overline{a_d^{pr}} a_d^{pr} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_q^p, m_d^r] - \\
& \frac{3}{8} \overline{a_d^{pr}} a_d^{pr} (g_1^2 + g_2^2) (2 g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_q^p, m_d^r] + \frac{1}{8} \overline{a_d^{pr}} a_d^{pr} (3 g_1^2 - g_2^2) \\
& (g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_q^p, m_d^r] - \frac{1}{2} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,0}[m_q^p, m_q^s] + \\
& \frac{3}{4} \overline{y_d^{pr}} \overline{y_d^{st}} y_d^{pt} y_d^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_q^p, m_q^s] - \frac{1}{48} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^4 - 18 g_1^2 g_2^2 - 3 g_2^4) \\
& \text{LF}_{2,2,0}[m_q^p, m_u^r] + \frac{1}{48} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^4 - 42 g_1^2 g_2^2 - 27 g_2^4) \text{LF}_{3,1,0}[m_q^p, m_u^r] + \\
& \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_q^p, m_u^r] + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^2 + g_2^2) \\
& \text{LF}_{4,1,-1}[m_q^p, m_u^r] + \frac{1}{48} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^4 - 18 g_1^2 g_2^2 - 3 g_2^4) \text{LF}_{3,1,0}[m_u^r, m_q^p] - \\
& \frac{1}{16} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,2,-1}[m_u^r, m_q^p] (g_1^2 + g_2^2)^2 + \frac{3}{16} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} \text{LF}_{4,1,-1}[m_u^r, m_q^p] (g_1^2 + g_2^2)^2 + \\
& \frac{1}{8} \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (3 g_1^2 - g_2^2) (g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_u^r, m_q^p] - \frac{1}{8} g_1^2 \text{LF}_{3,1,-1}[\tilde{\mu}, m_1] (g_1^2 + g_2^2)^2 + \\
& \frac{3}{16} g_1^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] + \frac{3}{16} g_1^4 m_1^2 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} g_1^2 (g_1^2 + g_2^2) (6 g_1^2 + g_2^2) \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{8} g_1^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] - \\
& \frac{1}{8} g_1^4 m_1^2 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{24} (-3 g_1^6 - 2 g_1^4 g_2^2 + g_1^2 g_2^4) \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} (-9 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + 7 g_2^6) \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] + \\
& \frac{85}{48} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] + \frac{3}{16} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\
& \frac{3}{8} (2 g_1^4 g_2^2 + g_1^2 g_2^4 - g_2^6) \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] - \frac{7}{8} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-3}[\tilde{\mu}, m_2] - \\
& \frac{1}{8} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{8} (-3 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + g_2^6) \text{LF}_{5,1,-3}[\tilde{\mu}, m_2] + \\
& g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{3}{4} m_1 m_2 g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& \frac{1}{2} g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] - \frac{1}{2} m_1 m_2 g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& \overline{y_d^{pr}} \overline{y_d^{st}} \overline{y_d^{uv}} y_d^{pt} y_d^{sv} y_d^{ur} \text{LF}_{1,1,1,0}[m_d^r, m_d^t, m_d^v] - \frac{1}{2} g_1^2 \overline{y_d^{st}} y_d^{sr} \overline{a_d^{pr}} a_d^{pr} \\
& \text{LF}_{2,1,1,0}[m_d^r, m_d^t, m_q^p] - \frac{1}{2} g_1^2 \overline{y_d^{st}} y_d^{sr} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{2,1,1,0}[m_d^t, m_d^r, m_q^p] + \frac{1}{3} \overline{y_e^{pr}} \overline{y_e^{st}} \overline{y_e^{uv}} y_e^{pt} y_e^{sv} y_e^{ur} \\
& \text{LF}_{1,1,1,0}[m_e^r, m_e^t, m_e^v] - \frac{1}{2} g_1^2 \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{2,1,1,0}[m_e^r, m_e^t, m_l^p] - \\
& \frac{1}{2} g_1^2 \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} \text{LF}_{2,1,1,0}[m_e^r, m_l^p, m_l^s] - \frac{1}{2} g_1^2 \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pr} \\
& \text{LF}_{2,1,1,0}[m_e^t, m_e^r, m_l^p] - \frac{1}{2} g_2^2 \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{2,1,1,0}[m_l^p, m_e^r, m_e^t] + \\
& \frac{1}{4} \overline{y_e^{st}} y_e^{sr} \overline{a_e^{pr}} a_e^{pr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_l^p, m_e^r, m_e^t] - \frac{1}{2} g_2^2 \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} \\
& \text{LF}_{2,1,1,0}[m_l^p, m_e^r, m_l^s] + \frac{1}{4} \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_l^p, m_e^r, m_l^s] + \\
& \frac{1}{3} \overline{y_e^{pr}} \overline{y_e^{st}} \overline{y_e^{uv}} y_e^{pt} y_e^{sv} y_e^{ur} \text{LF}_{1,1,1,0}[m_l^p, m_l^s, m_l^u] - \frac{1}{2} g_2^2 \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} \\
& \text{LF}_{2,1,1,0}[m_l^s, m_e^r, m_l^p] + \frac{1}{4} \overline{y_e^{st}} y_e^{pt} \overline{a_e^{pr}} a_e^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_l^s, m_e^r, m_l^p] - \\
& \frac{1}{2} \overline{y_d^{st}} y_d^{sr} \overline{a_d^{pr}} a_d^{pr} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^p, m_d^r, m_d^t] + \\
& \frac{3}{4} \overline{y_d^{st}} y_d^{sr} \overline{a_d^{pr}} a_d^{pr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^p, m_d^r, m_d^t] - \\
& \frac{1}{2} \overline{y_d^{st}} y_d^{pt} \overline{a_d^{pr}} a_d^{sr} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^p, m_d^r, m_q^s] + \\
& \frac{3}{4} \overline{y_d^{st}} y_d^{pt} \overline{a_d^{pr}} a_d^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^p, m_d^r, m_q^s] + \\
& \overline{y_d^{pr}} \overline{y_d^{st}} \overline{y_d^{uv}} y_d^{pt} y_d^{sv} y_d^{ur} \text{LF}_{1,1,1,0}[m_q^p, m_q^s, m_q^u] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) \text{LF}_{2,1,1,0}[m_q^p, m_q^s, m_u^t] - \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^p, m_q^s, m_u^t] - \\
& \frac{3}{8} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_q^p, m_u^t, m_q^s] - \\
& \frac{1}{2} \overline{y_d^{st}} y_d^{pt} \overline{a_d^{pr}} a_d^{sr} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^s, m_d^r, m_q^p] + \\
& \frac{3}{4} \overline{y_d^{st}} y_d^{pt} \overline{a_d^{pr}} a_d^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^s, m_d^r, m_q^p] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) \text{LF}_{2,1,1,0}[m_q^s, m_q^p, m_u^t] - \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^s, m_q^p, m_u^t] - \\
& \frac{3}{8} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_q^s, m_u^t, m_q^p] - \\
& \frac{3}{8} g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] + \frac{7}{8} g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + \\
& \frac{3}{8} m_1 m_2 g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] - \frac{1}{4} g_1^4 g_2^2 \text{LF}_{3,2,1,-3}[\tilde{\mu}, m_1, m_2] - \\
& \frac{1}{4} m_1 g_1^4 g_2^2 (m_1 + 2 m_2) \text{LF}_{3,2,1,-2}[\tilde{\mu}, m_1, m_2] - \frac{1}{2} g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] - \\
& \frac{1}{4} m_1 m_2 g_1^2 g_2^2 (g_1^2 + g_2^2) \text{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \frac{1}{4} g_1^2 g_2^2 (2 g_1^2 + 3 g_2^2) \text{LF}_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] - \\
& \frac{1}{4} m_2 g_1^2 g_2^2 (2 m_1 g_1^2 + g_2^2 (4 m_1 + m_2)) \text{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] + \\
& 3 \overline{y_d^{st}} \overline{y_d^{uv}} y_d^{sv} y_d^{ur} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{1,1,1,1,0}[m_d^r, m_d^t, m_d^v, m_q^p] - \\
& \frac{1}{2} g_1^2 \overline{a_d^{pr}} \overline{a_d^{st}} a_d^{pt} a_d^{sr} \text{LF}_{2,1,1,1,0}[m_d^r, m_d^t, m_q^p, m_q^s] + \\
& 3 \overline{y_d^{st}} \overline{y_d^{uv}} y_d^{pt} y_d^{ur} \overline{a_d^{pr}} a_d^{sv} \text{LF}_{1,1,1,1,0}[m_d^r, m_d^t, m_q^p, m_q^s] + \\
& 3 \overline{y_d^{st}} \overline{y_d^{uv}} y_d^{pv} y_d^{ut} \overline{a_d^{pr}} a_d^{sr} \text{LF}_{1,1,1,1,0}[m_d^r, m_q^p, m_q^s, m_q^u] + \\
& \overline{y_e^{st}} \overline{y_e^{uv}} y_e^{sv} y_e^{ur} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{1,1,1,1,0}[m_e^r, m_e^t, m_e^v, m_l^p] - \\
& \frac{1}{2} g_1^2 \overline{a_e^{pr}} \overline{a_e^{st}} a_e^{pt} a_e^{sr} \text{LF}_{2,1,1,1,0}[m_e^r, m_e^t, m_l^p, m_l^s] + \\
& \overline{y_e^{st}} \overline{y_e^{uv}} y_e^{pt} y_e^{ur} \overline{a_e^{pr}} a_e^{sv} \text{LF}_{1,1,1,1,0}[m_e^r, m_e^v, m_l^p, m_l^s] + \\
& \overline{y_e^{st}} \overline{y_e^{uv}} y_e^{pv} y_e^{ut} \overline{a_e^{pr}} a_e^{sr} \text{LF}_{1,1,1,1,0}[m_e^r, m_l^p, m_l^s, m_l^u] - \\
& \frac{1}{2} g_2^2 \overline{a_e^{pr}} \overline{a_e^{st}} a_e^{pt} a_e^{sr} \text{LF}_{2,1,1,1,0}[m_l^p, m_e^r, m_e^t, m_l^s] + \\
& \frac{1}{4} \overline{a_e^{pr}} \overline{a_e^{st}} a_e^{pt} a_e^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,1,-1}[m_l^p, m_e^r, m_e^t, m_l^s] - \\
& \frac{1}{2} \overline{a_d^{pr}} \overline{a_d^{st}} a_d^{pt} a_d^{sr} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,1,0}[m_q^p, m_d^r, m_d^t, m_q^s] + \\
& \frac{3}{4} \overline{a_d^{pr}} \overline{a_d^{st}} a_d^{pt} a_d^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,1,-1}[m_q^p, m_d^r, m_d^t, m_q^s] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_u^{st}} y_u^{pt} \overline{a_d^{pr}} a_d^{sr} (g_1^2 + g_2^2) \text{LF}_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^r, m_u^t] - \\
& g_1^2 \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \text{LF}_{2,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t] + \\
& \frac{3}{4} \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,1,-1}[m_q^p, m_q^s, m_u^r, m_u^t] + \\
& \frac{3}{8} \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + g_2^2) \text{LF}_{2,2,1,1,-1}[m_q^p, m_u^r, m_q^s, m_u^t] + \\
& \frac{3}{8} \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + g_2^2) \text{LF}_{2,2,1,1,-1}[m_q^p, m_u^t, m_q^s, m_u^r] + \\
& \frac{1}{4} \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,1,0}[m_u^r, m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{4} \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + g_2^2) \text{LF}_{3,1,1,1,-1}[m_u^r, m_q^p, m_q^s, m_u^t] + \\
& \frac{3}{8} \tilde{\mu}^4 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + g_2^2) \text{LF}_{2,2,1,1,-1}[m_u^r, m_u^t, m_q^p, m_q^s] + \\
& 3 \overline{y_d^{uv}} y_d^{ur} \overline{a_d^{pr}} \overline{a_d^{st}} a_d^{pt} a_d^{sv} \text{LF}_{1,1,1,1,1,0}[m_d^r, m_d^t, m_d^v, m_q^p, m_q^s] + \\
& 3 \overline{y_d^{uv}} y_d^{sv} \overline{a_d^{pr}} \overline{a_d^{st}} a_d^{pt} a_d^{ur} \text{LF}_{1,1,1,1,1,0}[m_d^r, m_d^t, m_q^p, m_q^s, m_q^u] + \\
& \overline{y_e^{uv}} y_e^{ur} \overline{a_e^{pr}} \overline{a_e^{st}} a_e^{pt} a_e^{sv} \text{LF}_{1,1,1,1,1,0}[m_e^r, m_e^t, m_e^v, m_l^p, m_l^s] + \\
& \overline{y_e^{uv}} y_e^{sv} \overline{a_e^{pr}} \overline{a_e^{st}} a_e^{pt} a_e^{ur} \text{LF}_{1,1,1,1,1,0}[m_e^r, m_e^t, m_l^p, m_l^s, m_l^u] + \\
& \overline{a_d^{pr}} \overline{a_d^{st}} \overline{a_d^{uv}} a_d^{pt} a_d^{sv} a_d^{ur} \text{LF}_{1,1,1,1,1,1,0}[m_d^r, m_d^t, m_d^v, m_q^p, m_q^s, m_q^u] + \\
& \frac{1}{3} \overline{a_e^{pr}} \overline{a_e^{st}} \overline{a_e^{uv}} a_e^{pt} a_e^{sv} a_e^{ur} \text{LF}_{1,1,1,1,1,1,0}[m_e^r, m_e^t, m_e^v, m_l^p, m_l^s, m_l^u] + \\
& \tilde{\mu}^6 \overline{y_u^{pr}} \overline{y_u^{st}} \overline{y_u^{uv}} y_u^{pt} y_u^{sv} y_u^{ur} \text{LF}_{1,1,1,1,1,1,0}[m_q^p, m_q^s, m_q^u, m_u^r, m_u^t, m_u^v] \Big)
\end{aligned}$$